

Sunwoo Kim

PhD Student — Compiler-aided reasoning for chip design

Ithaca, NY | sk3463@cornell.edu | (607) 262-0961 | sunwookim028.github.io | github.com/sunwookim028

Education

-
- Cornell University**, PhD in Computer Engineering Aug 2025 – present
- Advisor: Prof. Zhiru Zhang
 - Selected coursework: Advanced Compilers; High-Level Digital Design Automation
- Seoul National University**, BS in Electrical & Computer Engineering (cum laude) Mar 2019 – Feb 2025

Experience

-
- Student Researcher**, Computer Systems Lab, Cornell University – Ithaca, NY Aug 2025 – present
- Research: Chip design infrastructure using MLIR-based abstractions and LLM agents (see Projects)
 - Mentoring: 14 undergraduate researchers across two semesters; several since interned at AMD and Micron
 - Service: Artifact Evaluation Committee, MLSys 2026
- Teaching Assistant**, School of ECE, Cornell University – Ithaca, NY Jan 2026 – present
- Developing hardware design and programming labs for a new ECE course on AI hardware, taught by Prof. Zhang
- Research Intern**, Accelerated Intelligent Systems Lab, SNU – Seoul, Korea Jan 2024 – June 2025
- Co-led kernel design and indexing optimization for G³SA, a GPU-accelerated sequence alignment library
 - Co-first author on ACM ICS 2025 publication

Projects

-
- Compiler-Aided Reasoning for Agentic RTL Design** ([link](#)) Sept 2025 – present
- Compiler infrastructure and composability as grounding for LLM-based hardware design.*
- Implemented a parametrized TPU in Allo; extended Allo with scratchpad modeling and kernel-launch programming to support ASIC beyond FPGA backends (~15× fewer lines of code than our Verilog baseline)
 - Compiler-based composition of HLS and IP modules for verifiable assembly, targeting a CGRA design
 - MLIR-based microarchitecture abstractions above HLS/RTL for automated ISA extraction and compiler backend generation, targeting the TPU design
 - Task decomposition and position-independent context caching for large-scale design tasks
 - Harness design and post-training for EDA-specific tool use and reasoning
- Agentic PyTorch Backend for Heterogeneous Acceleration** ([link](#)) Oct 2025 – present
- Agent-driven backend adaptation for evolving heterogeneous hardware (e.g., edge NPUs, chiplet-based SiPs) and kernel libraries.*
- Program partitioning (e.g., PD disagg.) and compilation for heterogeneous systems (NPU+GPU, CGRA+PIM)
 - Demonstrated on AMD Strix Halo with IRON and ROCm libraries
 - Automated kernel-library registration and partitioning-logic updates via LLM agents
- Hardware-Accelerated Scalable Tooling** ([link](#)) Jan 2024 – present
- Multi-GPU software for genomics; GPU-accelerated simulation for chip design.*
- Multi-GPU acceleration for high-throughput NGS pipelines (kernels for all seed, chain, extend alignment stages)
 - Triton migration for long-term maintainability
 - GPU-accelerated architecture simulation for chip design

Selected Honors

- Fulbright Award for Graduate Studies in STEM (graciously declined), \$40k (2024)
- Korea Presidential Science Scholarship, full tuition and stipend (2019–2024)
- SNU Semiconductor Specialization Scholarship, \$11k (2023)

Publications

G³SA: A GPU-Accelerated Gold Standard Genomics Library for End-to-End Sequence Alignment ([link](#)) ICS 2025

Yeejoo Han*, **Sunwoo Kim***, Seongyeon Park, Jinho Lee (* *equal contribution*)

From Pragmas to Partners: A Symbiotic Evolution of Agentic High-Level Synthesis LATTE 2026 (Workshop)
([link](#))

Niansong Zhang, **Sunwoo Kim**, Shreesha Srinath, Zhiru Zhang

Skills

Programming Languages: C, C++, CUDA, Python, Bash, Verilog, SystemVerilog

Frameworks & Tools: MLIR, PyTorch, Allo, Vitis HLS, cocotb, Vivado, SPICE, Make, gdb, git